

What Kind of Language is Difficult for Transformers? Revisiting Cross-Linguistic Difficulty Rankings in Language Modeling

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Abstract

Cross-linguistic language-model evaluation is difficult because apparent model performance depends not only on linguistic structure, but also on tokenization, writing system, training data, and the metric used for comparison. This project revisits [Mielke et al. \(2019\)](#), which studied language difficulty with RNN language models. We adapt their framework to a pretrained multilingual Transformer setting by evaluating mGPT on 20 languages from FLORES+, a fully parallel multilingual benchmark. For each aligned sentence, we compute sentence-level surprisal and compare several aggregate measures. Our results show that language difficulty rankings vary substantially across metrics, especially for languages using non-Latin scripts. Among the tested predictors, tokenizer fertility measured as tokens per character is the only feature significantly correlated with estimated difficulty. These findings suggest that, for multilingual Transformers, cross-linguistic difficulty is shaped less by corpus level traits and more by the interaction between model architecture, pretraining distribution, and tokenizer efficiency.

1 Introduction and Motivation

What kind of language is hard for language models? [Mielke et al. \(2019\)](#) argue that raw perplexity is not a reliable way to compare difficulty across languages because it depends heavily on the unit of prediction. They propose a cross-linguistic evaluation framework based on sentence-level surprisal, allowing model difficulty to be comparable across languages with different token inventories and scripts. They evaluate RNN language models on parallel multilingual data and asks whether language modeling difficulty can be explained by linguistic and typological properties, including WALS features, average dependency length, and other measures of structural complexity. They found that many intuitive explanations for language diffi-

culty have limited predictive power, suggesting language difficulty is not simply an inherent property of a language, but an interaction between linguistic structure, data representation, and model behavior.

Our project revisits this question in the context of pretrained multilingual Transformers since they are now more widely used. Transformers differ from RNNs in their use of self-attention, shared multilingual tokenization, and large-scale pretraining, so what makes a language appear difficult may also change. We adapt their framework by evaluating a Transformer language model on a fully parallel multilingual benchmark. We will compute sentence-level surprisal and compare aggregate measures. Our goal is to test whether language difficulty factors changes with model architecture, tokenizer behavior, and pretraining distribution.

Our main hypothesis is that cross-linguistic difficulty in multilingual Transformers is not determined by linguistic structure alone. Instead, we hypothesize that tokenizer fertility will be a stronger predictor of mGPT difficulty. If this hypothesis is supported, it would suggest that the bottleneck in multilingual language modeling has shifted toward representing languages efficiently within a shared subword vocabulary.

The rest of this paper proceeds as follows. Section 2 reviews prior work on cross-linguistic language-model difficulty, multilingual Transformers, and tokenizer quality. Section 3 describes our experimental setup. Section 4 presents our results. Section 5 concludes by discussing what these findings suggest about language difficulty, representation efficiency, and fairer multilingual evaluation.

2 Related Works

Our project builds most directly on work that studies whether some languages are harder for language models than others. [Cotterell et al. \(2018\)](#) ask whether languages are equally difficult to model

by comparing n-gram and LSTM language models on translated text. They find that languages differ in modeling difficulty and suggest that inflectional morphology can contribute to these differences. Mielke et al. (2019) extend this line of work, arguing that raw perplexity is not a reliable measure for comparing languages because it depends on tokenization, writing system, and the unit of prediction. They use parallel multilingual text and introduce a multiplicative mixed-effects model that separates sentence-level content difficulty from language-level difficulty. Their results show that many intuitive linguistic predictors do not strongly explain RNN language modeling difficulty while surface-level corpus statistics do.

The move from RNNs to Transformers is central to our study. RNN language models process tokens sequentially, so information must be carried through recurrent hidden states. In contrast, Transformers use self-attention to model relationships between tokens more directly (Vaswani et al., 2017). This architectural difference may change what makes a language difficult for a model. Transformers also introduce new sources of variation, especially through large-scale multilingual pretraining and shared subword tokenization. Rust et al. (2021) show that multilingual model performance is strongly affected by tokenizer quality: languages that are poorly represented in a shared vocabulary are often split into more subword units, which can hurt performance. This connects to our analysis of tokenizer fertility, or how many model tokens are produced per character or byte. If a language is heavily fragmented into subword units, the model must make more prediction steps for the same sentence content, increasing apparent difficulty.

3 Method

We adapt the evaluation framework of Mielke et al. (2019) to a modern pretrained multilingual Transformer setting, evaluating mGPT (Shliazhko et al., 2022) on FLORES+ (NLLB Team et al., 2024) dataset. We compute sentence-level surprisal and estimate language difficulty using multiple metrics. We also conduct correlation analysis to identify which features predict difficulty.

We use mGPT as the primary language model. It is a multilingual autoregressive Transformer language model, making it suitable for computing sentence-level surprisal directly from next-token probabilities. It uses shared parameters across lan-

guage and is pretrained on large-scale multilingual text.

We use FLORES+ (NLLB Team et al., 2024) as the primary evaluation corpus. It is a fully-parallel multilingual benchmark containing 3,001 sentences per language, allowing us to compare sentence-level surprisal across languages while holding the underlying sentence intent approximately constant. The mGPT version we use was released in 2022, whereas FLORES+ was updated in 2024, reducing the likelihood of benchmark contamination.

20 languages, available in both mGPT and FLORES+, are selected across 17 language families to conduct our experiment.

Language Family	Chosen Languages
Afro-Asiatic	Arabic (ar), Hebrew (he)
Austro-Asiatic	Vietnamese (vi)
Austronesian	Indonesian (id), Javanese (jv)
Basque	Basque (eu)
Dravidian	Tamil (ta)
Indo-European (Germanic)	English (en), German (de)
Indo-European (Romance)	French (fr)
Indo-European (Greek)	Greek (el)
Indo-European (Indo-Aryan)	Hindi (hi)
Indo-European (Iranian)	Persian (fa)
Japonic	Japanese (ja)
Koreanic	Korean (ko)
Niger-Congo	Swahili (sw)
Slavic	Russian (ru)
Sino-Tibetan	Burmese (my)
Turkic (Oghuz)	Turkish (tr)
Uralic	Finnish (fi)

Table 1: Chosen languages grouped by language family.

3.1 Metrics

We compute several measures of language difficulty. Let x_{ij} denote the sentence in language j corresponding to aligned sentence intent i . After tokenizing x_{ij} with the mGPT tokenizer, we obtain a token sequence $x_{ij} = (w_{ij1}, w_{ij2}, \dots, w_{ijT_{ij}})$, where T_{ij} is the number of mGPT tokens in the sentence. The sentence-level surprisal is computed as the sum of token-level negative log probabilities: $S_{ij} = -\sum_{t=1}^{T_{ij}} \log p_{\theta}(w_{ijt} | w_{ij<t})$. A higher value of S_{ij} means that the model assigns a lower probability to the sentence, so the sentence is harder for the model to predict.

The first normalized measure is the average negative log likelihood per token:

$$\text{NLL}_{\text{token}}(j) = \frac{\sum_i S_{ij}}{\sum_i T_{ij}}$$

This metric measures how surprising each mGPT token is on average. This is highly sensitive to tokenization. If one language is split into many short subword tokens while another language is

represented using fewer larger tokens, token-level scores may not be directly comparable.

The second measure is the surprisal normalized by UTF-8 byte length. Let B_{ij} be the number of UTF-8 bytes in sentence x_{ij} . We define Bits Per Byte as

$$\text{BPB}(j) = \frac{\sum_i S_{ij}}{\ln(2) \sum_i B_{ij}}$$

This metric is more comparable across tokenizers because it does not directly depend on the number of subword tokens. However, it is still affected by writing system because different scripts require different numbers of bytes per character in UTF-8.

The third measure is the surprisal normalized by character length. Let C_{ij} be the number of characters in sentence x_{ij} . We compute Bits Per Character as

$$\text{BPC}(j) = \frac{\sum_i S_{ij}}{\ln(2) \sum_i C_{ij}}.$$

This avoids some byte-level encoding artifacts, but it is still not a perfect measure because characters are not equivalent across writing systems. A single written character may correspond to different linguistic units depending on the script.

Finally, we use the heteroscedastic multiplicative model adapted from Mielke et al. (2019) to estimate language-level difficulty while controlling for aligned sentence content. Let y_{ij} be the observed surprisal for sentence intent i in language j . The model assumes that surprisal is generated by a sentence-level difficulty term n_i , a language-level difficulty coefficient d_j , and residual noise:

$$y_{ij} = n_i \cdot \exp(d_j) \cdot \exp(\epsilon_{ij})$$

Taking logs gives

$$\log y_{ij} = \log n_i + d_j + \epsilon_{ij}$$

n_i captures how difficult the sentence content is across languages, while d_j captures how difficult language j is for the model after controlling for sentence content. We use English as the reference language. A positive d_j means that language j is harder than English for mGPT, while a negative d_j means that it is easier.

The heteroscedastic model allows the variance of the residuals to depend on sentence-level information. Following the original framework, this prevents long or highly informative sentences from dominating the difficulty estimates. Overall, the model gives a more controlled estimate of language-specific difficulty than raw token-, byte-, or character-normalized metrics.

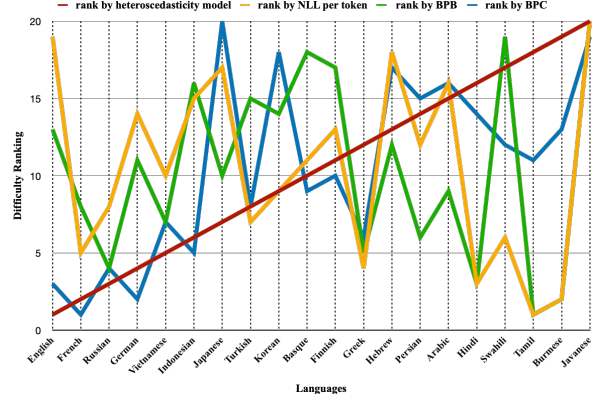


Figure 1: Difficulty rankings produced by different metrics.

3.2 Correlation Analysis

Following Mielke et al. (2019), we test six candidate features against the language difficulty coefficients d_j . The first three are computed directly from the FLORES+ test data: word inventory size, raw character sequence length, and raw byte sequence length. Two features capture tokenizer behavior: tokenizer fertility measured as mGPT tokens per byte and tokens per character. These two features were not tested in the original study, but they quantify how aggressively the shared multilingual tokenizer segments each language. Average dependency length is computed following Futrell et al. (2015) using the Stanza dependency parser (Qi et al., 2020). The original study computed this feature using UDPipe 1.2.0 (Straka et al., 2016), while we use Stanza as a modern alternative. Dependency length could not be computed for Swahili, Javanese, Burmese due to the absence of Stanza parsing models for these languages, reducing the sample size to 17 for this feature.

For each feature, we computed both Spearman’s rank correlation ρ with d_j , as it only requires a monotonic relationship and is more robust to outliers. Since we test six features simultaneously, we use Benjamini-Hochberg correction (Benjamini and Hochberg, 1995) for p-values to control the false discovery rate. We consider a feature significant only if its corrected p-value is ≤ 0.05 , following the threshold used in the original study.

4 Experiment and Result

4.1 Difficulty Ranking

Following the statistical filtering of surface-level artifacts via our heteroscedastic mixed-effect model, latent parameter representing the true intrinsic dif-

Tier	Relative Difficulty	Languages in Order ($\prec$)
Tier 1	Analytic / High-Resource	English $\prec$ French $\prec$ Russian $\prec$ German $\prec$ Vietnamese $\prec$ Indonesian $\prec$ Japanese $\prec$ Turkish
Tier 2	Morphologically Complex	Korean $\prec$ Basque $\prec$ Finnish $\prec$ Greek $\prec$ Hebrew $\prec$ Persian $\prec$ Arabic
Tier 3	Low-Resource Non-Latin	Hindi $\prec$ Swahili $\prec$ Tamil $\prec$ Burmese $\prec$ Javanese

Table 2: Calibrated latent language difficulty hierarchy

difficulty are successfully isolated for each target language. The results are as shown in figure 1.

4.1.1 Resource Scale and Typological Vulnerability

The calibrated model validated several assumptions regarding resource allocation and metric stability in multilingual NLPs.

English as the optimal baseline English consistently exhibits the lowest intrinsic difficulty as the standardized baseline. This is an expected outcome driven by the architectures of Transformer and English-centric pretraining data, leading to optimal token alignment.

Resource dominance A strong correlation is observed between severe data scarcity and high intrinsic difficulty. Low-resources that employ non-Latin scripts tend to be systematically penalized by models, as they lack the token efficiency found in high-resource baselines.

Cross-level metric vulnerability As demonstrated in figure 1, traditional metrics shift violently depending on whether cross-linguistic evaluation is conducted at the token, byte, or character level. This variance confirms that unadjusted benchmarks measure surface-level data choices rather than actual structural complexity.

4.1.2 Granular Analysis of the Intrinsic Difficulty Ranking

By controlling surface-level noise, our model established the underlying difficulty ranking for the evaluated languages.

Latent Difficulty Hierarchy Sorted in ascending order from the most sample-efficient (easiest) to the most parameter-taxing (hardest), the hierarchy is formulated in table 2.

The high-resource tier consists primarily of isolating or fusional languages backed by massive

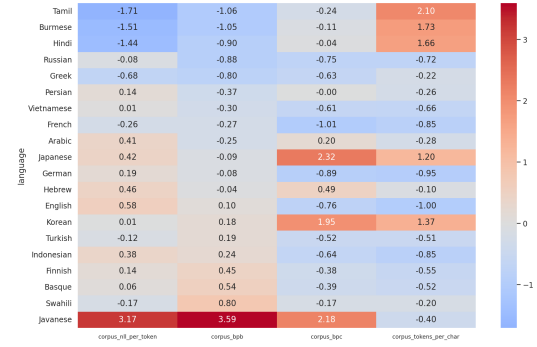


Figure 2: Normalized cross-Lingual metrics

pretraining data. They are able to be processed without much effort because their syntactic relations are expressed through rigid word orders or highly frequent inflections that the model can easily memorize.

The morphologically complex tier is dominated by agglutinative languages (e.g., Turkish, Finnish, Basque) and non-concatenative root-and-pattern languages (e.g., Arabic, Hebrew). A single word-form in these languages may contain multiple stacked morphemes. Because subword tokenizers frequently fracture these complex words into unaligned pieces, the model faces higher uncertainty per token.

The low-resource non-Latin tier occupies the hardest position in the ranking. Languages in this tier suffer from a compounding disadvantage of their highly complex morphosyntactic structures and severe under-representation in mGPT pretraining corpus.

Byte-inflation illusion in non-Latin scripts The most surprising finding centers on the failure of naive metrics when evaluating non-Latin scripts like Tamil, Burmese, and Hindi. As shown in figure 1, unadjusted metrics like surprisal per token and surprisal per UTF-8 byte falsely categorize Tamil, Burmese, and Hindi as the easiest languages, ranking them near the bottom of raw difficulty scales.

This is a major statistical illusion. Because non-Latin scripts require up to 3 bytes per character (as shown in Figure 2), their raw text files contain an artificially inflated number of bytes, causing inflation of denominators when computing information-theoretic metrics. After our model isolates and discards this surface noise, Tamil and Burmese are revealed among the top 3 most difficult languages in the corpus.

The Modern Transformer and Traditional RNN Paradox In traditional studies utilizing RNNs or

LSTMs, morphosyntactically complex languages like German and Russian were ranked as the most difficult languages. However, as seen in Figure 1, under modern mGPT transformer setup, German and Russian collapse entirely into the highly efficient zone, clustering tightly with English. This suggests that in the modern LLM era, pretraining data volume overpowers typological syntax complexity.

4.2 Correlating Factors

Table 3 summarizes the Spearman rank correlations between each candidate feature and the difficulty coefficients d_j . Of the six features tested, only tokenizer fertility in tokens per character is significant after Benjamini-Hochberg correction ($\rho = 0.758$, $p_{\text{corr}} < 0.001$). Languages where mGPT’s tokenizer produces more tokens per character are consistently harder to model. The remaining five features show no significant correlation.

This result sharply contrasts with Mielke et al. (2019), where word inventory size and raw character sequence length were the dominant predictors of difficulty for RNNLMs. Neither feature shows any correlation under mGPT. This change may be explained by two factors. Word inventory size matters in the original study because RNNLMs were trained on small corpora where large-vocabulary languages lack sufficient data per subword unit. mGPT is pretrained on billions of tokens, so it no longer faces this bottleneck. Moreover, character sequence length matters because RNNs struggle with long sequences due to information decay. Transformers handle this through self-attention.

Feature	N	ρ	p_{corr}
Word Inventory Size	20	0.033	0.890
Char Sequence Length	20	-0.060	0.890
Byte Sequence Length	20	0.289	0.434
Fertility (tokens/byte)	20	0.403	0.234
Fertility (tokens/char)	20	0.758	<0.001
Avg Dependency Length	17	0.189	0.702

Table 3: Spearman rank correlation with d_j . P-values corrected via Benjamini-Hochberg ($\alpha = 0.05$).

5 Conclusion

This study investigates the drives of cross-lingual difficulty in modern transformer-based language models, specifically focusing on mGPT. Extending the framework of Mielke et al. (2019), we applied Heteroscedastic translation difficulty model to the

parallel FLORES+ dataset across 20 diverse languages.

The empirical and statistical analysis yielded three primary conclusions. First, tokenizer fertility is identified as the single significant predictor of language difficulty for transformers, while traditional predictors for RNNLMs showed no significant correlation. Second, data scale overpowers grammar complexity, where low-resource languages exhibited higher difficulty, demonstrating that massive pretraining data scale and self-attention effectively overcomes traditional syntactic bottlenecks. Third, raw information-theoretic metrics are fundamentally unstable and causes byte-inflation illusion. Our findings indicate that linguistic difficulty is not an inherent property of a language’s grammar, but is architecture and data dependent.

Several questions remain open. It is unclear whether our findings generalize beyond mGPT or to a larger set of languages. The primary bottleneck in multilingual modeling has shifted from syntactic processing to representation efficiency. Future research could evaluate byte-level transformers to completely eliminate tokenization as a confounding variable, train smaller models from scratch on balanced dataset to fully disentangle the effects of data volume from architectural bias, or expand the language scope to include more low-resource language families across alternative open-source LLMs.

Ultimately, ensuring equitable cross-lingual performance requires the NLP community to pivot toward more efficient and localized tokenization strategies.

6 Contribution Statement

Shuhang Dong contributes to the implementation and interpretation of correlation analysis. Lyanna Chen contributes to the implementation of computing sentence-level surprisal and various difficulty measures for rankings. Shumin Yan contributes to the analysis of difficulty rankings and various results.

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